MG2DSCECT100 - ESSENTIAL CONCEPTS IN DIGITAL ELECTRONICS (2024 ADMISSION ONWARDS)

1. [c] 16

2. [c] 4

3. [a] 101100

4. [c] 111

5. [c] Both A and B

6. [a] -25

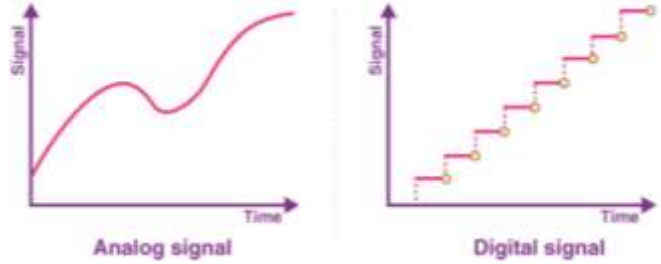
7. [b] 1001

8. [b] (A+B)(A+B+C)

9. [a] AND

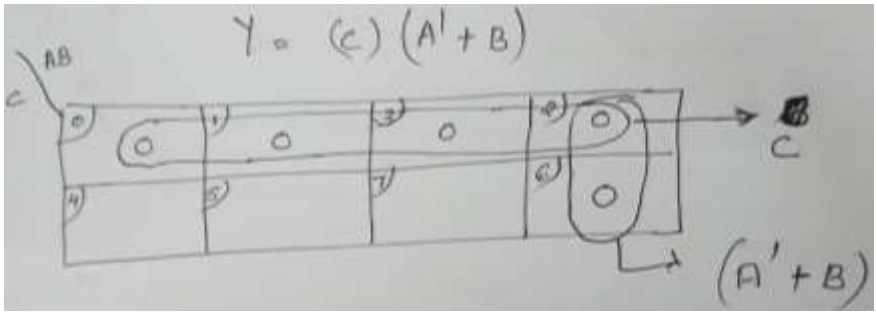
10. [b] 0000

11. Analog signals - continuous signals / Represented by sine waves / Human voice, natural sound are few examples / Continuous range of values / only used in analog devices.
Digital signals - Discrete signals / Represented by square wave / Output from computers, optical devices are few examples / Discontinuous values / Suited for digital electronics like computer, mobiles, embedded systems etc,.

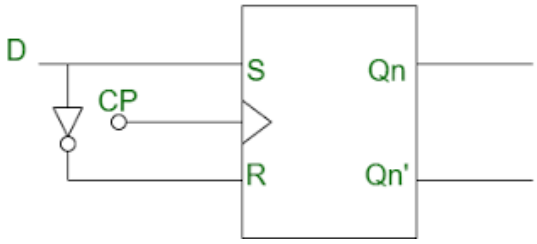


Give full mark for similar explanation. Otherwise give partial marks as per the content.

12. Give step by step marks
Plotting K Map for three variable system – 1 Mark
Mapping given Boolean expression to K Map – 1 Mark
Grouping 0' in the K Map – 1 Mark
For proper naming of groups – 1 Marks
Final simplified expression – 1 Mark



13. The D flip-flop has one data input (D) and one clock input (CLK). At the active edge of the clock (usually the rising edge), the logic value on D is transferred to the output Q (Q_{next} = D). The output Q remains unchanged Q_{prev} until the next active clock edge, regardless of changes to D in the meantime, making it ideal for storing a single bit of data. It is typically implemented by adding an inverter between the inputs of an SR flip-flop (connecting S=D and R=D', ensuring they are always opposite.



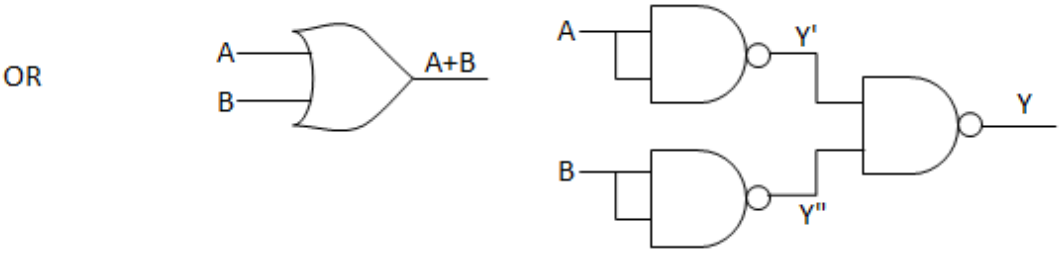
Truth Table

Q	D	$Q_{(t+1)}$
0	0	0
0	1	1
1	0	0
1	1	1

Similar diagram & Truth table 1 mark each. Proper explanation 3marks

14.

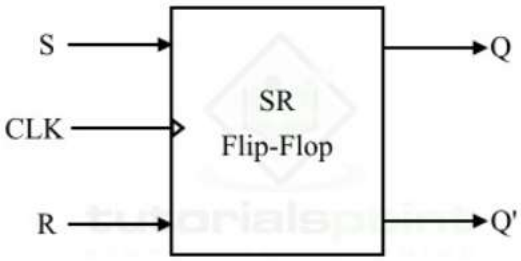
Logic Function Only **Symbol** **Circuit using NAND gate**



Similar diagram 3 marks, Explanation with Truth Table 2 Marks

15. Explanation with truth table and Logical diagram give 4 +1 = 5 Marks. Otherwise partial marks

Inputs			Output	Comment
S	R	Q_n	Q_{n+1}	
0	0	0	0	No Change / Hold
0	0	1	1	No Change / Hold
0	1	0	0	Reset
0	1	1	0	Reset
1	0	0	1	Set
1	0	1	1	Set
1	1	0	X	Forbidden
1	1	1	X	Forbidden



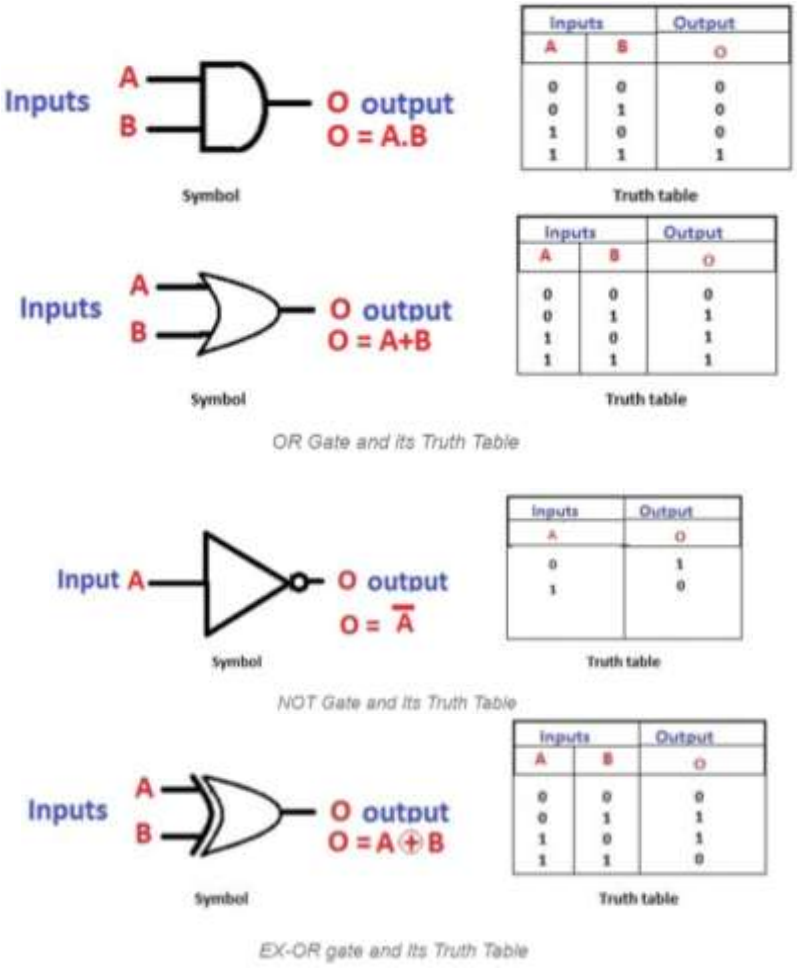
16. [a] To convert 103 to binary, divide by 2 and record the remainders from bottom to top.
103 div 2 = 51 with a remainder of 1 (LSB)
51 div 2 = 25 with a remainder of 1
25 div 2 = 12with a remainder of 1
12 div 2 = 6 with a remainder of 0
6 div 2 = 3 with a remainder of 0
3 div 2 = 1 with a remainder of 1
1 div 2 = 0 with a remainder of 1 (MSB)
Reading the remainders from bottom to top, 103 in binary is 1100111_2.

[b] To convert 86 to binary, divide by 2 and record the remainders from bottom to top.
86 div 2 = 43 with a remainder of 0 (LSB)
43 div 2 = 21 with a remainder of 1
21 div 2 = 10 with a remainder of 1
10 div 2 = 5 with a remainder of 0
5 div 2 = 2 with a remainder of 1
2 div 2 = 1 with a remainder of 0
1 div 2 = 0 with a remainder of 1 (MSB)

Reading the remainders from bottom to top, 86 in binary is 1010110_2

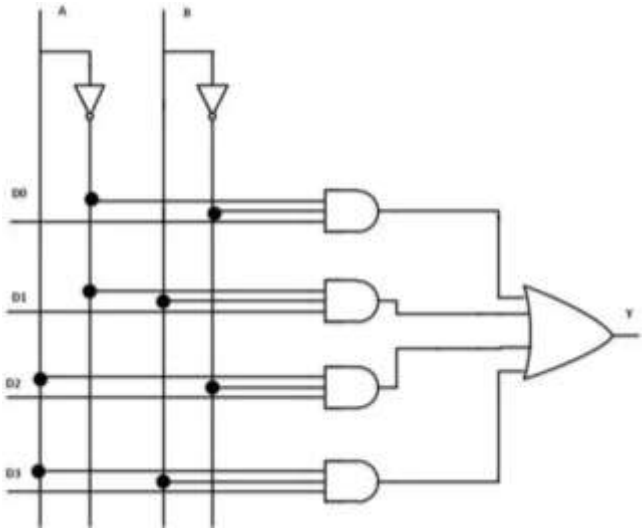
Give full marks if answer is correct, with steps clearly shown. Only answer 1+1 = 2 Marks

17. The AND gate is a digital logic gate with n inputs and one output, which performs logical conjunction based on the combinations of its inputs. The output of this gate is true only when all the inputs are true.



Give 2.5 Marks each for [a][b][c] and [d] parts

18.



X_1	X_0	Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Similar diagram 3 Marks, Similar Truth Table 2 Marks, Proper Explanation 5 Marks

19.

Name	AND form	OR form
Identity law	$1A = A$	$0 + A = A$
Null law	$0A = 0$	$1 + A = 1$
Idempotent law	$AA = A$	$A + A = A$
Inverse law	$A\overline{A} = 0$	$A + \overline{A} = 1$
Commutative law	$AB = BA$	$A + B = B + A$
Associative law	$(AB)C = A(BC)$	$(A + B) + C = A + (B + C)$
Distributive law	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
Absorption law	$A(A + B) = A$	$A + AB = A$
De Morgan's law	$\overline{AB} = \overline{A} + \overline{B}$	$\overline{A + B} = \overline{A}\,\overline{B}$

Write and describe any ten laws and rules from the above table. One mark each.

20. 5 marks for binary number system + 5 marks for hexadecimal number system.

The **binary number system** is a radix-2 number system with '0' and '1' as the two independent digits. All larger binary numbers are represented in terms of '0' and '1'. The procedure for writing higher order binary numbers after '1' is similar to the one explained in the case of the decimal number system. For example, the first 16 numbers in the binary number system would be 0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, 1100, 1101, 1110 and 1111. The next number after 1111 is 10000, which is the lowest binary number with five digits. This also proves the point made earlier that a maximum of only 16 (= 2^4) numbers could be written with four digits. Starting from the binary point, the place values of different digits in a mixed binary number are 2^0, 2^1 , 2^2 and so on (for the integer part) and 2^{-1} , 2^{-2} , 2^{-3} and so on (for the fractional part).

The **hexadecimal number system** is a radix-16 number system and its 16 basic digits are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F. The place values or weights of different digits in a mixed hexadecimal number are 16^0, 16^1, 16^2 and so on (for the integer part) and 16^{-1}, 16^{-2}, 16^{-3} and so on (for the fractional part). The decimal equivalent of A, B, C, D, E and F are 10, 11, 12, 13, 14 and 15 respectively. The hexadecimal number system provides a condensed way of representing large binary numbers stored and processed inside the computer